

R Markdown Part 2 Notes

Ashley Clark

2025-09-17

```
library(knitr)
opts_chunk$set(
  echo = FALSE, warning = FALSE, message = FALSE
)
```

RMarkdown Part 2

4.1 Installing Required Packages

4.2 Options for creating tables in R markdown

- Print a data.frame or matrix
 - Easy, but monospaced format
- Knitr::kable
 - Easy, nice looking – not 100% flexible
- Gt
 - Tidyverse compatible syntax, highly flexible, but only html for now
- Xtable:xtable
 - Highly customizable, but only for pdfs (LaTeX based)
- Pander::pander
 - Default tables for different R classes
- Stargazer::stargazer

4.3 Creating Table 1 using gtsummary

First, create new rmarkdown document

Descriptive Analysis

Tabular analysis

Characteristics of the virus from the CATNAP database are displayed in the table below.

Characteristic	Overall N = 611 ¹	PNGs in Glycan Shield		p-value ²
		< 4 N = 101 ¹	>= 4 N = 510 ¹	
Region				0.003
Asia	126 (21%)	29 (29%)	97 (19%)	
Europe.Americas	96 (16%)	5 (5.0%)	91 (18%)	
N.Africa	170 (28%)	26 (26%)	144 (28%)	
S.Africa	219 (36%)	41 (41%)	178 (35%)	
Length of Env protein	857 (850, 863)	852 (849, 859)	857 (851, 864)	<0.001
Antibiotic resistance	-0.27 (-0.76, 0.54)	0.24 (-0.62, 0.73)	-0.29 (-0.77, 0.36)	0.030

¹n (%); Median (Q1, Q3)

²Pearson's Chi-squared test; Wilcoxon rank sum test

Characteristic	Beta	95% CI	p-value
shield	-0.11	-0.19, -0.04	0.003
Region			<0.001
Asia	—	—	
Europe.Americas	-0.33	-0.57, -0.09	
N.Africa	-0.38	-0.59, -0.17	
S.Africa	-0.03	-0.23, 0.17	
Length of Env protein	0.02	0.01, 0.02	<0.001

Abbreviation: CI = Confidence Interval

4.5 Fitting and Summarizing Regression Model in R Markdown

Regression Analysis

Primary Model

Secondary model

In this analysis, we generated a binary outcome where level 1 of the outcome means the antibody resistance score was greater than 1.

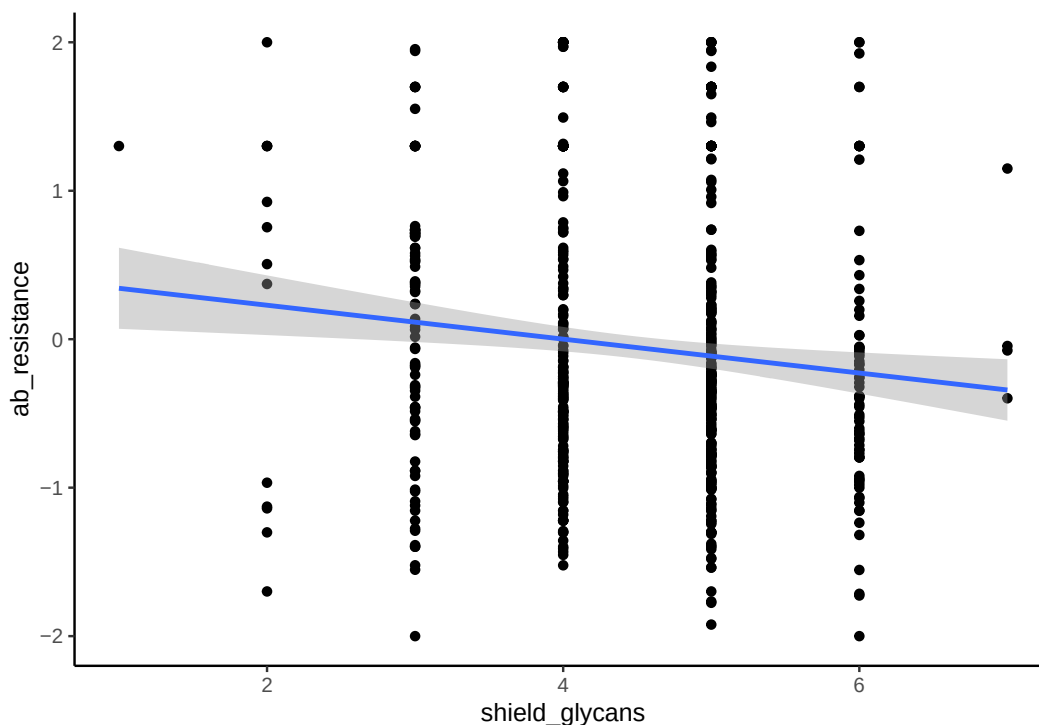
4.6 Adding figures to an R Markdown document

Graphical Analysis

Below we report a scatter plot of antibody resistance as a function of number of PNGs in the glycan shield.

Characteristic	OR	95% CI	p-value
shield	0.82	0.66, 1.03	0.082
Region			0.036
Asia	—	—	
Europe.Americas	0.76	0.34, 1.60	
N.Africa	0.84	0.44, 1.59	
S.Africa	1.63	0.93, 2.92	
Length of Env protein	1.06	1.04, 1.08	<0.001

Abbreviations: CI = Confidence Interval, OR = Odds Ratio



4.7 Setting Global Chunk Options

- Useful if you don't want code chunks to show in rendered rmarkdown document
- “Set Chunk Opts at Top”

4.8 Dynamically setting code chunk options

- production or test chunk at the top
 - creates flag if this report was for two different audiences

4.10 Parameterized R Markdown reports

1. new rmarkdown file?

option added to yaml header

“params:” “variable1: 10, variable2: 20, variable3: 30”

[1] "The value of variable 1 is..."

NULL